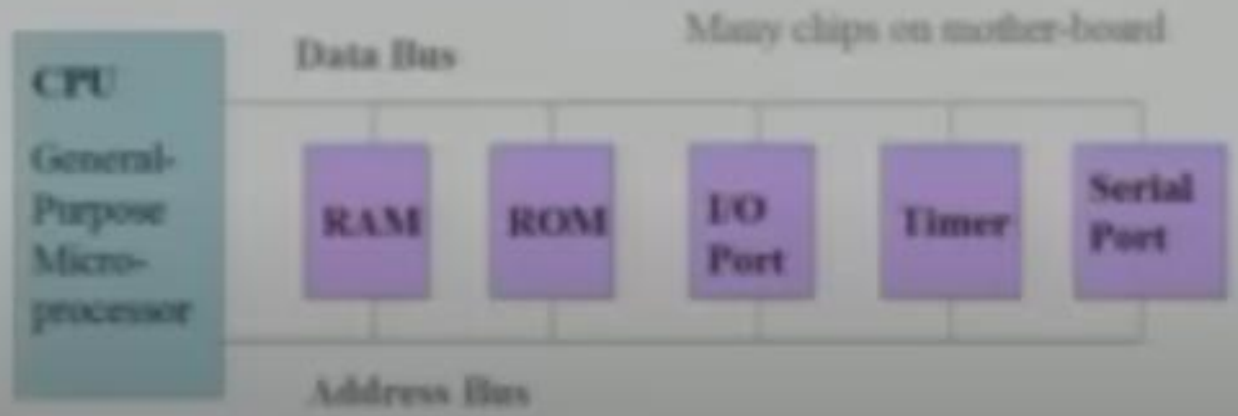
Architecture

https://mrcet.com/downloads/digital_notes/ECE/IV%20Year/10082021/ESD%20Digital%20Notes.pdf

Microprocessors

- CPU for Computers
- No RAM, ROM, I/O on CPU chip itself
- Example: Intel's x86, Motorola's 680x0

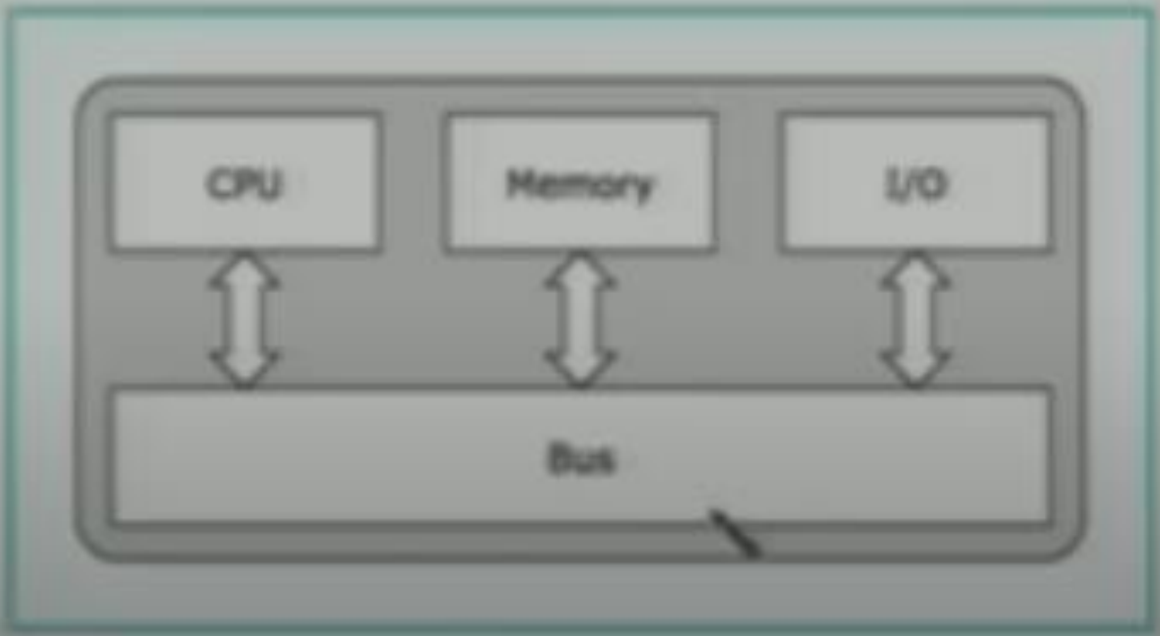


General-Purpose Microprocessor System

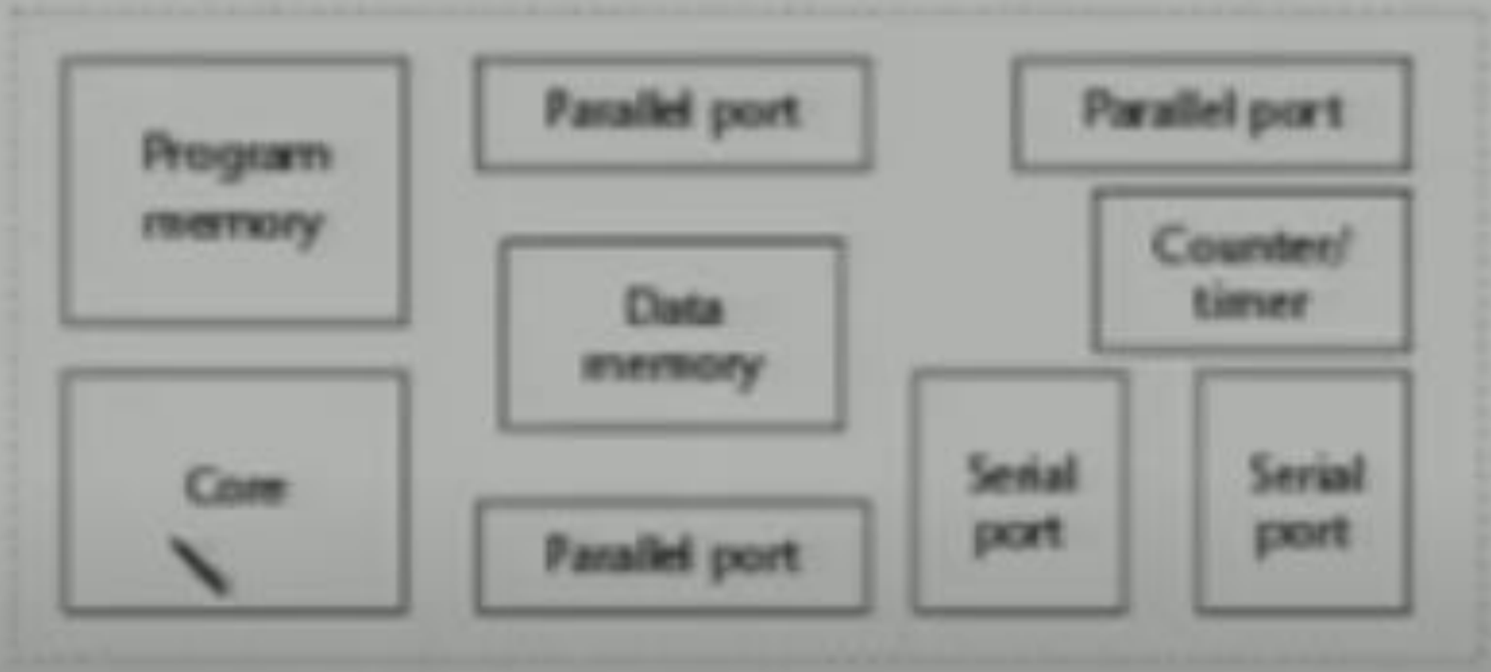
What is Micro-controller?

- Basically, a micro-controller is a device which integrates a number of the components of a microprocessor system onto a single microchip.
- A micro-controller combines onto the same microchip:
  - The CPU core
  - Memory (both ROM and RAM)
  - Some parallel digital I/O & *more*

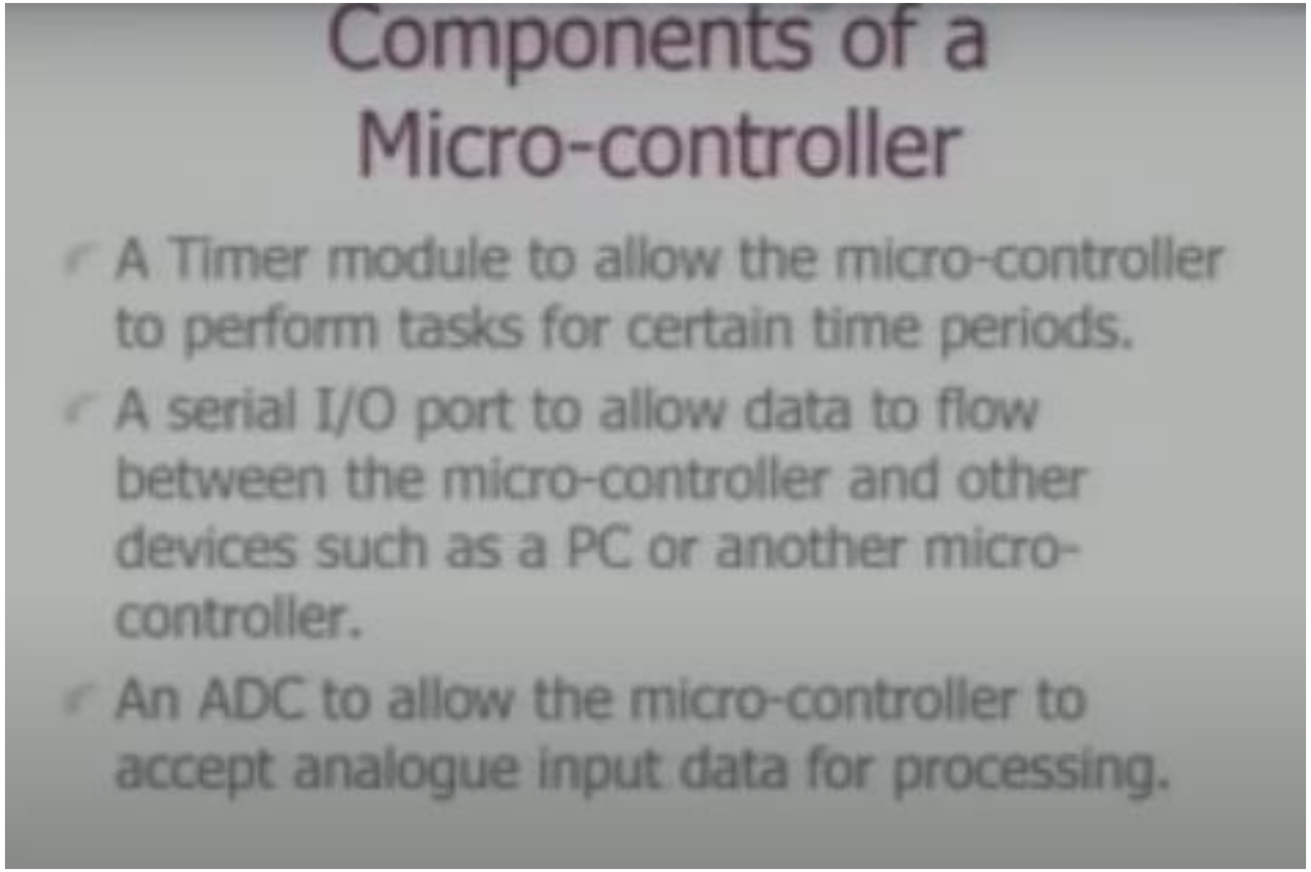
Micro-controller



Micro-controller

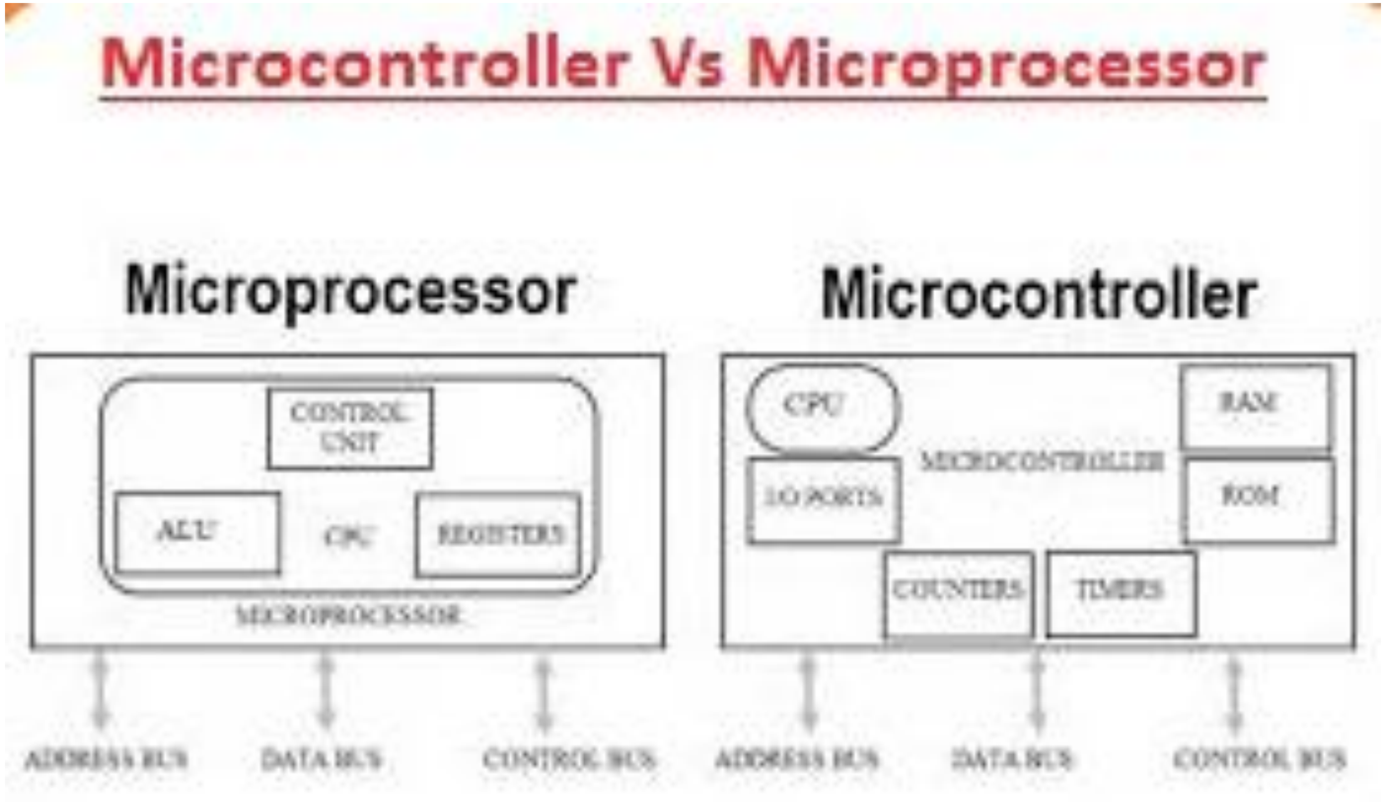


Certain time period, interrupt –switching from one task to another



Why Micro-controller?

- Low cost, small packaging
- Low power consumption
- Programmable, re-programmable
- Lots of I/O capabilities
- Easy integration with circuits
- For applications in which cost, power and space are critical
- Single-purpose


Microcontroller Vs Microprocessor
Microprocessor
Microcontroller
CONTROL UNIT
ALU
CPU
REGISTERS
MICROPROCESSOR
CPU
RAM
MICROCONTROLLER
I/O PORTS
ROM
COUNTERS
TIMERS
ADDRESS BUS
DATA BUS
CONTROL BUS
ADDRESS BUS
DATA BUS
CONTROL BUS

DSP vs. Microcontroller

- DSP
  - **Harvard Architecture**
  - VLIW/SIMD (parallel execution units)
  - No bit level operations
  - Hardware MACs
  - DSP applications
- Microcontroller
  - **Mostly von Neumann Architecture**
  - Single execution unit
  - Flexible bit-level operations
  - No hardware MACs
  - Control applications

VonNeuman Architecture

- Only one bus between CPU and memory
- RAM and program memory share the same bus and the same memory, and so must have the same bit width
- Bottleneck: Getting instructions interferes with accessing RAM

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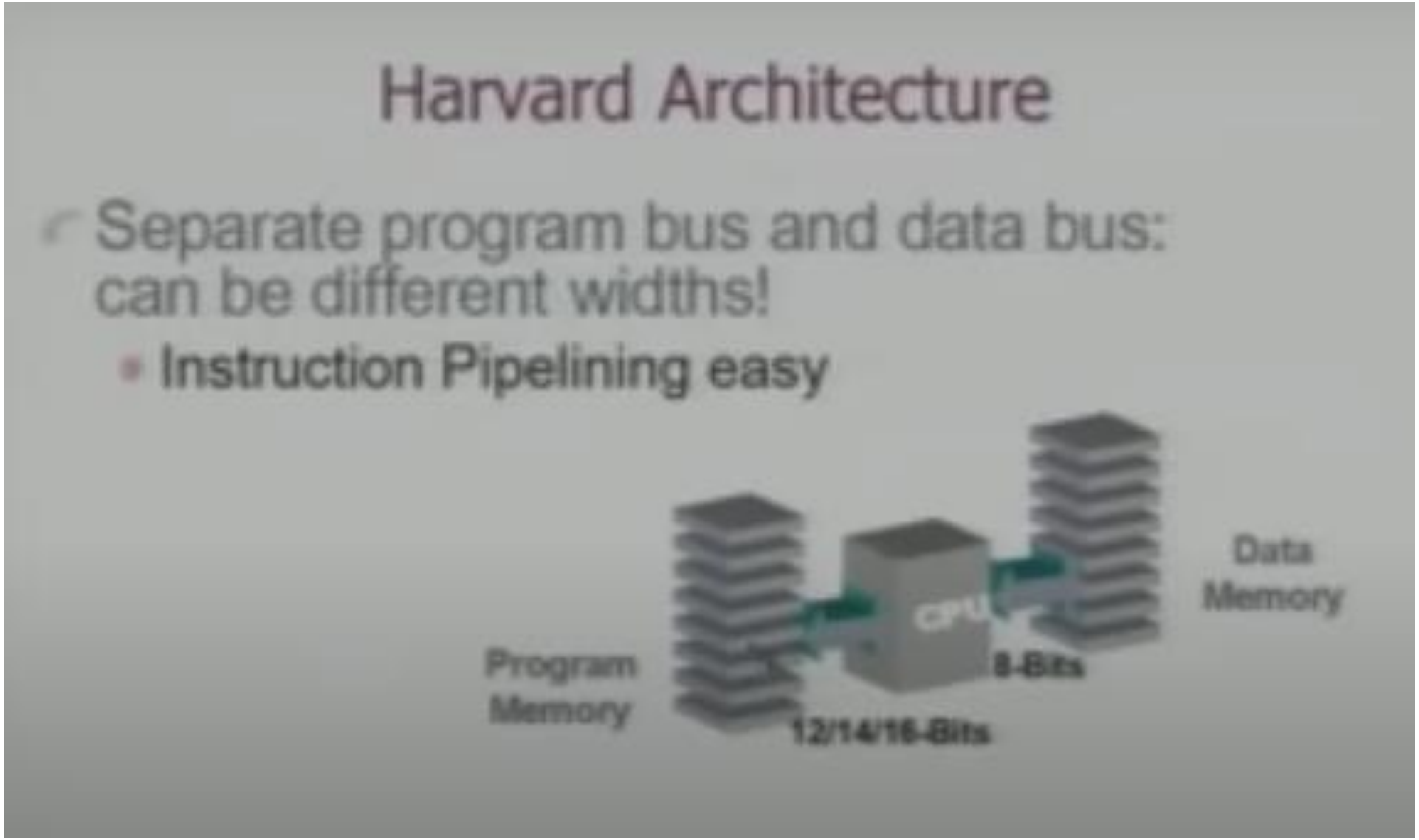
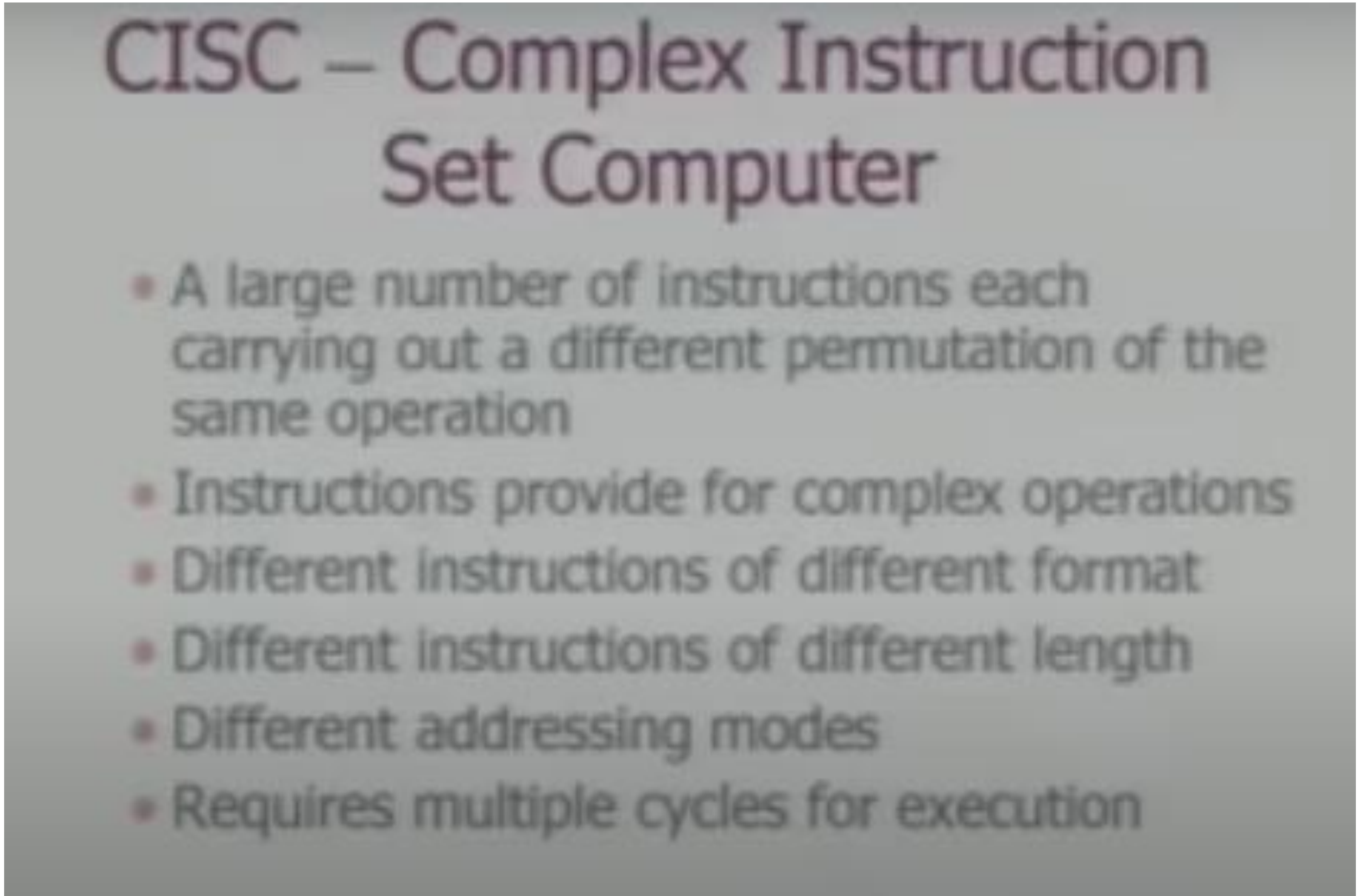

Harvard Architecture
Separate program bus and data bus: can be different widths!
Instruction Pipelining easy
Data Memory
CPU
8-Bits
Program Memory
12/14/16-Bits

Pentium processor, 8086,6800,complex task, lot of addressing modes,



RISC – Reduced Instruction Set Computer

- Instructions for simple operations that can be executed in a single cycle
- Each instruction of fixed length
  - Facilitates instruction pipelining
- Large general purpose register set
  - Can contain data or address (*symmetry*)
- Load-store Architecture
  - No memory access for data processing instructions

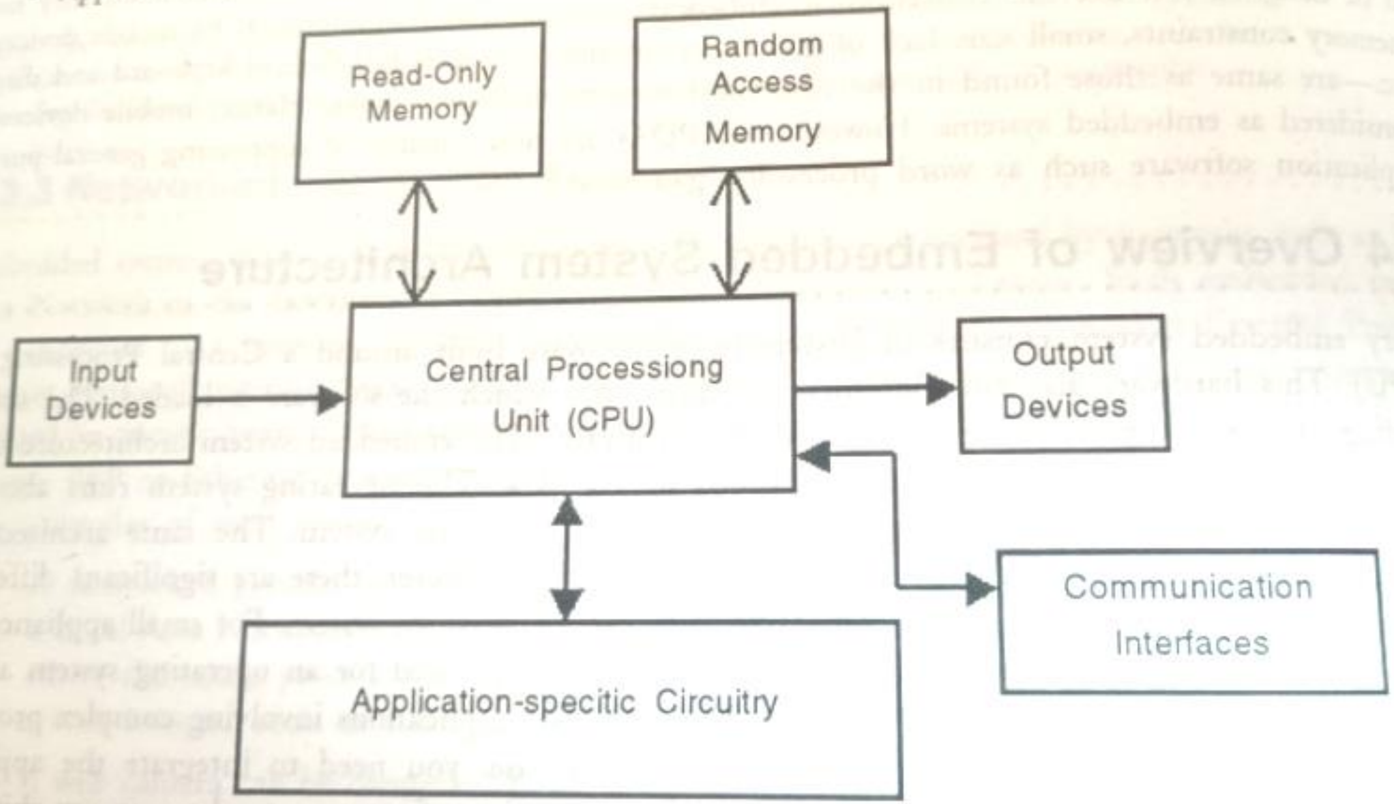
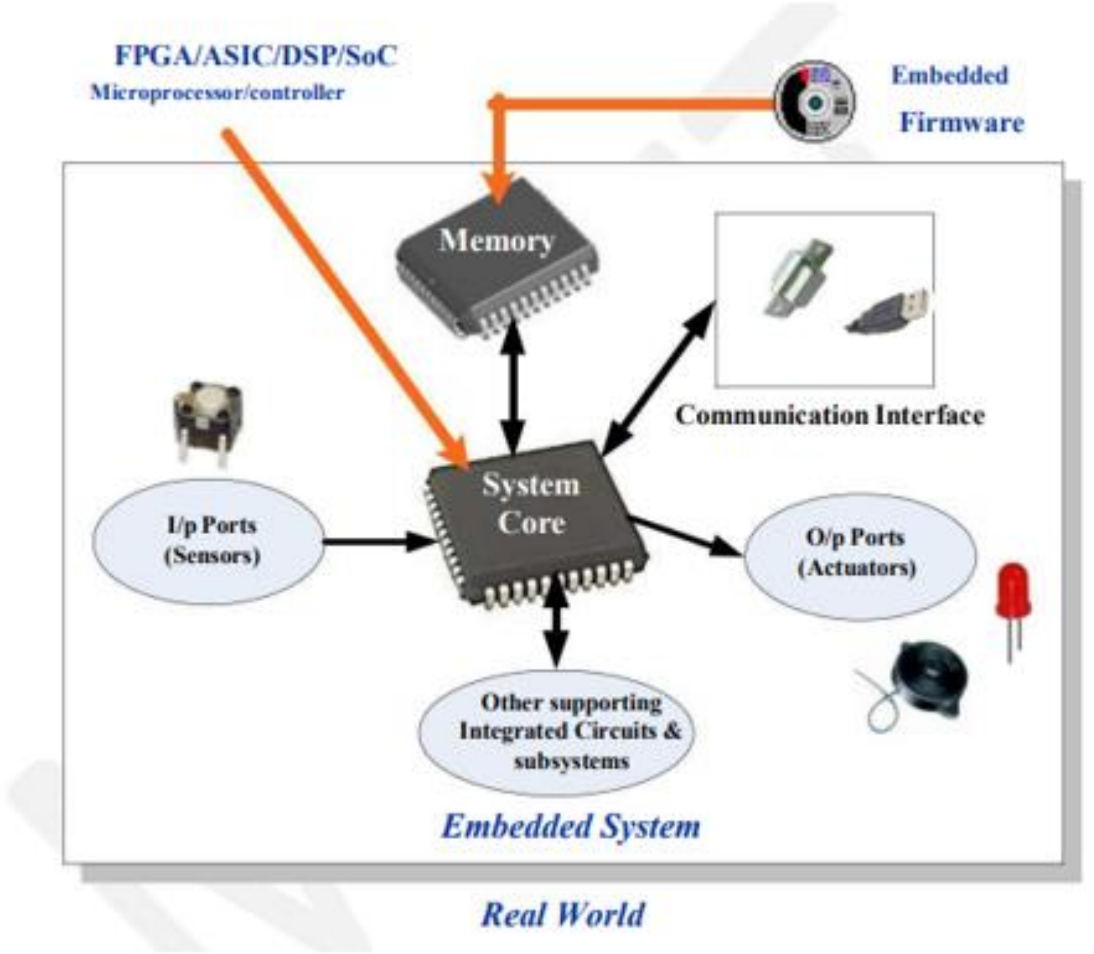


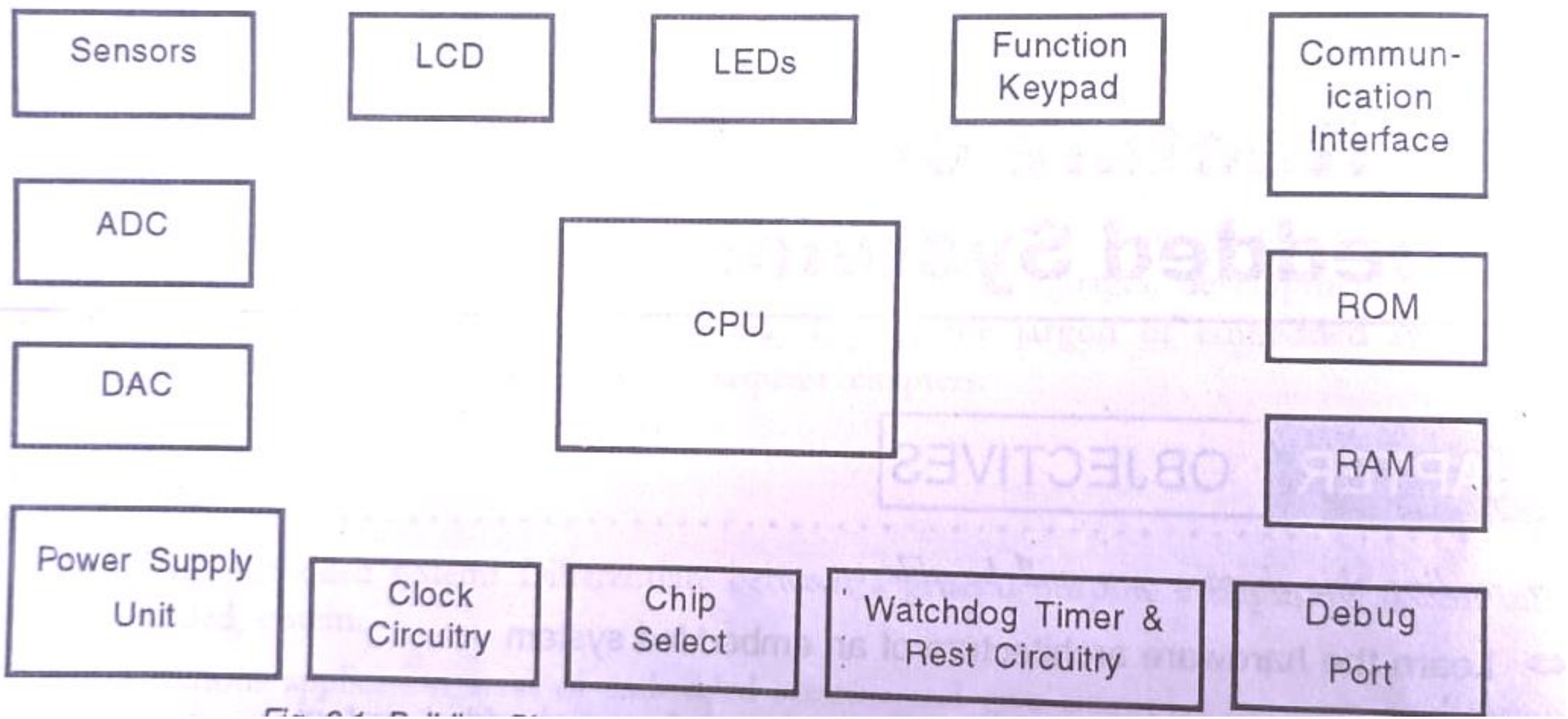
Fig. 1.5: Simplified Hardware Architecture of an Embedded System

- details of the various building blocks of the hardware of an embedded system. the building blocks are:
- Central Processing Unit (CPU)
- Memory (Read-only Memory and Random Access Memory)
- Input DevicesOutput devices
- Communication interfaces
- Application-specific circuitry



FPGA/ASIC/DSP/SoC
Microprocessor/controller
Embedded
Firmware
Memory
Communication Interface
System
Core
I/p Ports
(Sensors)
O/p Ports
(Actuators)
Other supporting
Integrated Circuits &
subsystems
Embedded System
Real World

building blocks of the hardware of an embedded system.



CPU

- Central Processing Unit (CPU): The Central Processing Unit (processor, in short) can be any of following:
- micro-controller,
- microprocessor or
- Digital Signal Processor (DSP).
- A micro-controller low-cost processor. Its main attraction is that on the chip itself, there will be many other components such as memory, serial communication interface, analog-to-digital converter etc. So, for small applications micro-controller is the best choice as the number of external components required will be very
- On the other hand, microprocessors are more powerful, but you need to use many external components with them.
- DSP is used mainly for applications in which signal processing is involved such as and video processing.

CPU

- The CPU consists of
- Arithmetic Logic Unit (ALU) which performs arithmetic and logic operations (such as add, multiply, subtract, etc.)
- General purpose registers. These registers constitute the processor's internal memory. The number of registers varies from processor to processor. Registers contain the current data and operands that are being manipulated by the processor. When a processor is referred to as of 8-bits, 16-bits etc., it refers to the width of the registers.
- Control unit that fetches the instructions from memory, decodes them and executes them. A control unit consists of
- Instruction Pointer that points to the next instruction to be executed. Instruction Pointer is also called Program Counter.
- Stack Pointer that points to the stack in the memory. In the external memory, the processor implements a stack. When required, the contents of registers are transferred to this stack. The processor keeps track of the next free location in the stack through stack pointer.
- Instruction Decoder that decodes the instructions.
- Memory Address Register and Memory Data Register.Processor Architectures

Program Architecture:

Memory:

- The memory is categorized as Random Access Memory (RAM) and Read Only Me (ROM). The contents of the RAM will be erased if power is switched off to the chip, whereas retains the contents even if the power is switched off. So, the firmware is stored in the ROM. power is switched on, the processor

Memory

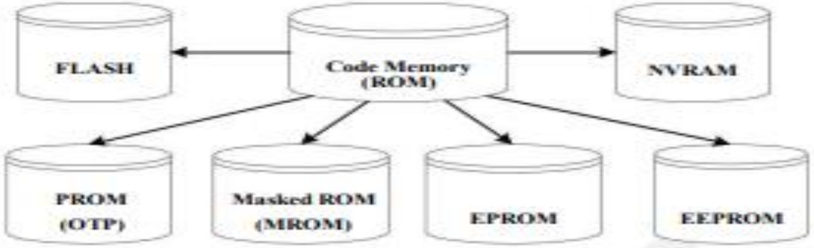
Memory:

- Memory is an important part of an embedded system. The memory used in embedded system can be either Program Storage Memory (ROM) or Data memory (RAM)
- Certain Embedded processors/controllers contain built in program memory and data memory and this memory is known as on-chip memory
- Certain Embedded processors/controllers do not contain sufficient memory inside the chip and requires external memory called **off-chip memory or external memory.**

Memory – Program Storage Memory:

- Stores the program instructions
- Retains its contents even after the power to it is turned off. It is generally known as Non volatile storage memory
- Depending on the fabrication, erasing and programming techniques they are classified into

Memory



1. **Masked ROM (MROM):**

- One-time programmable memory.
- Uses hardwired technology for storing data.
- The device is factory programmed by masking and metallization process according to the data provided by the end user.
- The primary advantage of MROM is low cost for high volume production.
- MROM is the least expensive type of solid state memory.
- Different mechanisms are used for the masking process of the ROM, like
  - Creation of an enhancement or depletion mode transistor through channel implant
  - By creating the memory cell either using a standard transistor or a high threshold transistor.
  - In the high threshold mode, the supply voltage required to turn ON the transistor is above the normal ROM IC operating voltage.
  - This ensures that the transistor is always off and the memory cell stores always logic 0.
- The limitation with MROM based firmware storage is the inability to modify the device firmware against firmware upgrades.
- The MROM is permanent in bit storage, it is not possible to alter the bit information

Memory

2. Programmable Read Only Memory (PROM) / (OTP) :

- It is not pre-programmed by the manufacturer
- The end user is responsible for Programming these devices.
- PROM/OTP has *nichrome* or *polysilicon* wires arranged in a matrix, these wires can be functionally viewed as fuses.
- It is programmed by a PROM programmer which selectively burns the fuses according to the bit pattern to be stored.
- Fuses which are not blown/burned represents a logic "1" where as fuses which are blown/burned represents a logic "0".The default state is logic "1".
- OTP is widely used for commercial production of embedded systems whose proto-typed versions are proven and the code is finalized.
- It is a low cost solution for commercial production.
- OTPs cannot be reprogrammed.

Memory

3. Erasable Programmable Read Only Memory (EPROM):

- Erasable Programmable Read Only (EPROM) memory gives the flexibility to re-program the same chip.
- During development phase , code is subject to continuous changes and using an OTP is not economical.
- EPROM stores the bit information by charging the floating gate of an FET
- Bit information is stored by using an EPROM Programmer, which applies high voltage to charge the floating gate
- EPROM contains a quartz crystal window for erasing the stored information. If the window is exposed to Ultra violet rays for a fixed duration, the entire memory will be erased
- Even though the EPROM chip is flexible in terms of re-programmability, it needs to be taken out of the circuit board and needs to be put in a UV eraser device for 20 to 30 minutes

Memory

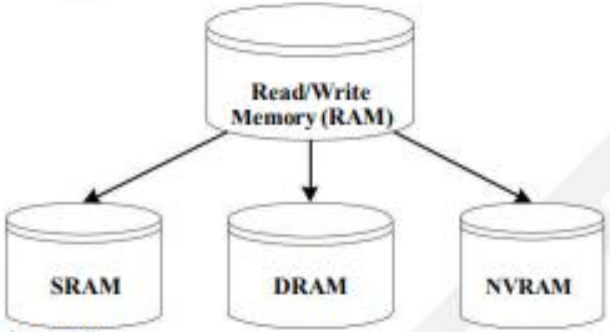
5. Program Storage Memory – FLASH

- FLASH memory is a variation of EEPROM technology.
- FALSH is the latest ROM technology and is the most popular ROM technology used in today"s embedded designs
- It combines the re-programmability of EEPROM and the high capacity of standard ROMs
- FLASH memory is organized as sectors (blocks) or pages
- FLASH memory stores information in an array of floating gate MOSFET transistors
- The erasing of memory can be done at sector level or page level without affecting the other sectors or pages
- Each sector/page should be erased before re-programming
- The typical erasable capacity of FLASH is of the order of a few 1000 cycles.

Memory:

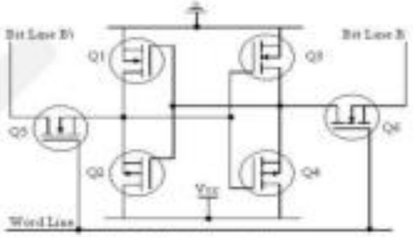
Read-Write Memory/Random Access Memory (RAM)

- ❖ RAM is the data memory or working memory of the controller/processor
- ❖ RAM is volatile, meaning when the power is turned off, all the contents are destroyed
- ❖ RAM is a direct access memory, meaning we can access the desired memory location directly without the need for traversing through the entire memory locations to reach the desired memory position (i.e. Random Access of memory location)



1. Static RAM (SRAM):

- ❖ Static RAM stores data in the form of Voltage.
- ❖ They are made up of flip-flops
- ❖ In typical implementation, an SRAM cell (bit) is realized using 6 transistors (or 6 MOSFETs).
- ❖ Four of the transistors are used for building the latch (flip-flop)part of the memory cell and 2 for controlling the access.
- ❖ Static RAM is the fastest form of RAM available.
- ❖ SRAM is fast in operation due to its resistive networking and switching capabilities

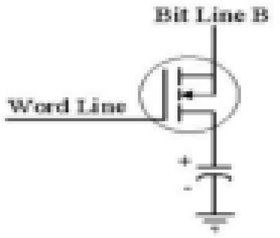


SRAM cell implementation

Memory

2. Dynamic RAM (DRAM)

- Dynamic RAM stores data in the form of charge. They are made up of MOS transistor gates
- The advantages of DRAM are its high density and low cost compared to SRAM
- The disadvantage is that since the information is stored as charge it gets leaked off with time and to prevent this they need to be refreshed periodically
- Special circuits called DRAM controllers are used for the refreshing operation. The refresh operation is done periodically in milliseconds interval



DRAM cell implementation

I/O devices & sensors/LCD/LEDs/ETC.

I/O /ETC.

Sensors & Actuators:

- Embedded system is in constant interaction with the real world
- Controlling/monitoring functions executed by the embedded system is achieved in accordance with the changes happening to the Real World.
- The changes in the system environment or variables are detected by the sensors connected to the input port of the embedded system.
- If the embedded system is designed for any controlling purpose, the system will produce some changes in controlling variable to bring the controlled variable to the desired value.
- It is achieved through an actuator connected to the out port of the embedded system.

Sensor:

- A transducer device which converts energy from one form to another for any measurement or control purpose. Sensors acts as input device
- Eg. Hall Effect Sensor which measures the distance between the cushion and magnet in the Smart Running shoes from adidas
- **Example: IR, humidity , PIR(passive infra red) , ultrasonic , piezoelectric , smoke sensors**

Soil Moisture Sensor Module

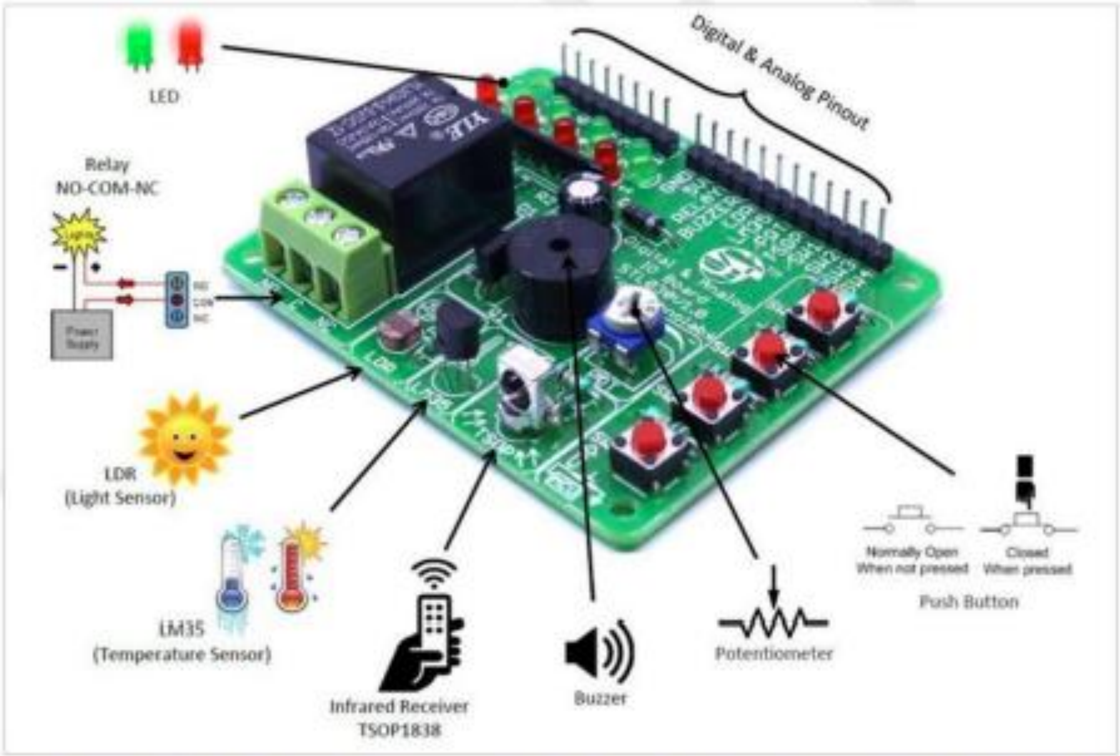
IR Infrared Obstacle Avoidance Sensor Module

ultrasonic sensor

I/O (s/ETC.

Actuator:

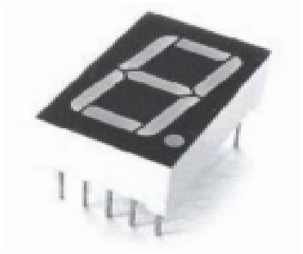
- A form of transducer device (mechanical or electrical) which converts signals to corresponding physical action (motion). Actuator acts as an output device
- Eg. Micro motor actuator which adjusts the position of the cushioning element in the Smart Running shoes from adidas



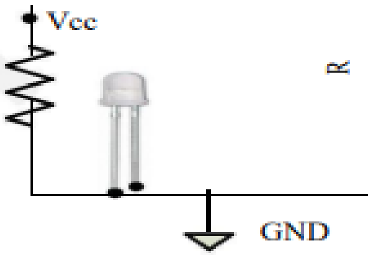
I/O devices & sensors/LCD/LEDs/ETC.

The I/O Subsystem:

- The I/O subsystem of the embedded system facilitates the interaction of the embedded system with external world
- The interaction happens through the sensors and actuators connected to the Input and output ports respectively of the embedded system
- The sensors may not be directly interfaced to the Input ports, instead they may be interfaced through signal conditioning and translating systems like ADC, Optocouplers etc



1. I/O Devices - Light Emitting Diode (LED):



- Light Emitting Diode (LED) is an output device for visual indication in any embedded system
- LED can be used as an indicator for the status of various signals or situations.
- Typical examples are indicating the presence of power conditions like „Device ON", „Battery low" or „Charging of battery" for a battery operated handheld embedded devices
- LED is a p-n junction diode and it contains an anode and a cathode.
- For proper functioning of the LED, the anode of it should be connected to +ve terminal of the supply voltage and cathode to the –ve terminal of supply voltage
- The current flowing through the LED must limited to a value below the maximum current that it can conduct.
- A resister is used in series between the power supply and the resistor to limit the current through the LED

I/O devices & sensors/LCD/LEDs/ETC.

2. I/O Devices – 7-Segment LED Display

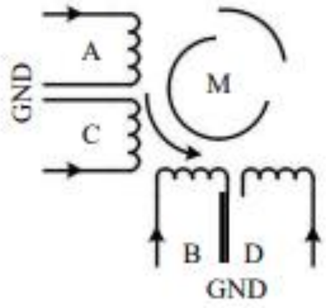
- The 7 – segment LED display is an output device for displaying alpha numeric characters
- It contains 8 light-emitting diode (LED) segments arranged in a special form. Out of the 8 LED segments, 7 are used for displaying alpha numeric characters

- The LED segments are named A to G and the decimal point LED segment is named as DP
- The LED Segments A to G and DP should be lit accordingly to display numbers and characters
- The 7 – segment LED displays are available in two different configurations, namely; Common anode and Common cathode
- In the Common anode configuration, the anodes of the 8 segments are connected commonly whereas in the Common cathode configuration, the 8 LED segments share a common cathode line
- Based on the configuration of the 7 – segment LED unit, the LED segment anode or cathode is connected to the Port of the processor/controller in the order „A" segment to the Least significant port Pin and DP segment to the most significant Port Pin.
- The current flow through each of the LED segments should be limited to the maximum value supported by the LED display unit

I/O devices & sensors/LCD/LEDs/ETC.

4. I/O Devices – Stepper Motor:

- Stepper motor is an electro mechanical device which generates discrete displacement (motion) in response to dc electrical signals
- It differs from the normal dc motor in its operation. The dc motor produces continuous rotation on applying dc voltage whereas a stepper motor produces discrete rotation in response to the dc voltage applied to it
- Stepper motors are widely used in industrial embedded applications, consumer electronic products and robotics control systems
- The paper feed mechanism of a printer/fax makes use of stepper motors for its functioning.
- Based on the coil winding arrangements, a two phase stepper motor is classified into



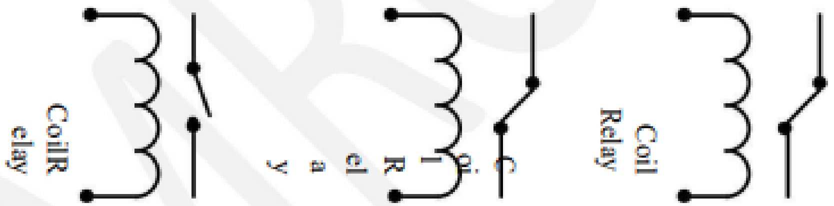
✓ Unipolar

✓ Bipolar

I/O devices & sensors/LCD/LEDs/ETC.

5. The I/O Subsystem – I/O Devices – Relay:

- An electro mechanical device which acts as dynamic path selectors for signals and power.
- The „Relay" unit contains a relay coil made up of insulated wire on a metal core and a metal armature with one or more contacts.
- „Relay" works on electromagnetic principle.
- When a voltage is applied to the relay coil, current flows through the coil, which in turn generates a magnetic field.



Single Pole Single Throw Normally Open | **Single Pole Single Throw Normally Closed** | **Single Pole Double Throw**

- The magnetic field attracts the armature core and moves the contact point.
- The movement of the contact point changes the power/signal flow path.
- The Relay is normally controlled using a relay driver circuit connected to the port pin of the processor/controller

I/O devices & sensors/LCD/LEDs/ETC.

6. The I/O Subsystem – I/O Devices -Piezo Buzzer:

- It is a piezoelectric device for generating audio indications in embedded applications.
- A Piezo buzzer contains a piezoelectric diaphragm which produces audible sound in response to the voltage applied to it.
- Piezoelectric buzzers are available in two types

1.Self-driving 2.External driving

- Self-driving contains are the necessary components to generate sound at a predefined tone.
- External driving piezo Buzzers supports the generation of different tones.
- The tone can be varied by applying a variable pulse train to the piezoelectric buzzer.
- A Piezo Buzzer can be directly interfaced to the port pin of the processor/Controller.

CLOCK CIRCUITRY or REAL-TIME CLOCK(RTC).

- The processor has to be given the clock input to one of the pins.
- To generate the clock signal, a crystal and oscillator are required. For some processors, the oscillator circuitry is in-built, only external crystal has to be added to generate the clock signal.
- Real Time Clock (RTC) keeps track of the date and time.
- All processor events are related to the clock. The higher the clock frequency, the higher the speed of the processor. However, different processors cannot be compared based on clock speed alone. One processor may take one clock cycle for executing an instruction, whereas another processor may take 16 clock cycles to execute the same instruction.
- RTC is used in embedded systems to provide "real" time including the date.

WATCHDOG TIMER

- In desktop Windows systems, if we feel our application is behaving in an abnormal way or if the system hangs up, we have the 'Ctrl + Alt + Del' to come out of the situation. What if it happens to our embed-ded system? Do we really have a 'Ctrl + Alt + Del't take control of the situation? Of course not, but we have a watchdog to monitor the firmware execution and reset the system processor/microcontroller when the program execution hangs up. A watchdog timer, or simply a watchdog, is a hardware timer for monitoring the firmware execution. Depending on the internal implementation, the watchdog timer in-crements or decrements a free running counter with each clock pulse and generates a reset signal to reset the processor if the count reaches zero for a down counting watchdog, or the highest count value for an upcounting watchdog. If the watchdog counter is in the enabled state, the firmware can write a zero (for upcounting watchdog implementation) to it before starting the execution of a piece of code (subroutine or portion of code which is susceptible to execution hang up) and the watchdog will start counting. If the firmware execution doesn't complete due to malfunctioning, within the time required by the watchdog to reach the maximum count, the counter will generate a reset pulse and this will reset the processor (if it is connected to the reset line of the processor). If the firmware execution completes before the expiration of the watchdog timer you can reset the count by writing a 0 (for an upcounting watchdog timer) to the watchdog timer register.

Input devices:

- Unlike the desktops, the input devices to an embedded system have very limited capability. There will be no keyboard or a mouse, and hence interacting with the embedded system is no easy task. Many embedded systems will have a small keypad you press one key to give a specific command. A keypad may be used to input only the digits. Many embedded systems used in process control do not have any input device for user interaction; they take inputs from sensors or transducers and produce electrical signals that are in turn fed to other systems.

Output devices:

- The output devices of the embedded systems also have very limited capability. Some embedded systems will have a few Light Emitting Diodes (LEDs) to indicate the health status of the system modules, or for visual indication of alarms. A small Liquid Crystal Display (LCD) may also be used to display some important parameters.

Communication interfaces:

- The embedded systems may need to interact with other embedded systems or they may have to transmit data to a desktop. To facilitate this, the embedded systems are provided with one or a few communication interfaces such as RS232, RS422, RS485, Universal Serial Bus (USB), IEEE 1394, Ethernet etc.

Application-specific circuitry:

- Sensors, transducers, special processing and control circuitry may be required for an embedded system, depending on its application. This circuitry interacts with the processor to carry out the necessary work.

The CPU of an embedded system can be

- : (a) micro-controller, (b) microprocessor, or (c) Digital Signal Processor.
- A micro-controller is the best choice for small embedded systems because it has memory and peripherals in the same chip as the CPU and hence very few extra components are required..
- Digital Signal Processors are the best choice if the application demands signal processing such as audio/video processing.

Specialities of Embedded Systems

- Developers need to keep these specialities in mind while designing embedded systems.

1. Reliability

- When we use a desktop, sometimes the system 'hangs' and we need to reset the computer. Generally, this does not cause any problem. However, this is not the case with the embedded systems used in mission-critical applications. They must work with high reliability

1. Reliability

- Reliability is of paramount importance in embedded systems. They should **continue** to work for thousands of hours without break.
- Many embedded systems used in industrial control are **inaccessible**. They are hidden in some other large-sized equipment; hence there will not be a reset button on such systems! So, the design of the embedded system should be such that in case the system has to be reset, the reset should be done thematically. Special hardware/software needs to be built into the system to take care of it. This special module is known as **watchdog timer.**
- Many embedded systems used in industrial automation and defence equipment need to work in **extreme environmental conditions such as very high/low temperatures, high humidity**. Besides, they should be able to withstand **bump and vibrations**. Hence, very stringent environmental specifications have to be met by such systems. The ability to work reliably in extreme environmental conditions is known as **ruggedness.** Not only the military equipment, even the consumer appliances such as the mobile phone need to be very rugged. Many people keep dropping their mobile phone on the floor,but still it works because it is very rugged

2 Performance

- Many embedded systems have time constraints. For instance, in a process control system, a constraint can be: "if the temperature exceeds 40 degrees, open a valve within 10 milliseconds." The system must meet such deadlines. If the deadlines are missed, it may result in a catastrophe. You can imagine the damage that can be done if such deadlines are not met in a safety system of a nuclear plant.

3 Power Consumption

- Most of the embedded systems operate through a battery. To reduce the battery drain and avoid frequent recharging of the battery, the power consumption of the embedded system has to be very low. To reduce power consumption such hardware components should be used that consume less power. Besides, emphasis should be on reducing the components count of the hardware. To reduce component count, the hardware designers have the option of using Programmable Logic Devices (PLDs) and Field Programmable Gate Arrays (FPGAs). Reducing the component count apart from reducing the power consumption also increases the reliability of the system.

4 Cost

- For embedded systems used in safety applications of a nuclear plant or in a spacecraft, cost may be a very important factor. However, for embedded systems used in consumer electronics or of automation, the cost is of utmost importance. Suppose you designed a toy in which the electro will cost US$20. By a careful analysis of the design, if you can find a way to reduce the cost to US it will be a great job. Don't underplay the importance of that one dollar because when you sel million toys, the cost reduction is $10 million!
- Not surprisingly, the hardware, engineers debat component selection to reduce even $0.1.

5. Size

- Size is certainly a factor for many embedded systems. We do not like a mobile phone that has to carried on our backs. The size and the weight are important parameters in embedded systems use in aircraft, spacecraft, missiles etc. because in such cases, every inch and every gram matters. reduce the size and the weight, again the hardware engineers have to design their boards by reducir the component count to the maximum possible extent. During the initial development of mobi phones based on GSM (Global System for Mobile Communication), GSM was expanded as "Go Send Mobiles" because of the complexity of the mobile phone. Today, the mobile phones go into o pockets, thanks to developments in microelectronics.